UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

5390878785

PHYSICS **5054/03**

Paper 3 Practical Test **May/June 2008**

2 hours

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
For each of the questions in Section A, you will be allowed to work with the apparatus for a maximum of 20 minutes. For the question in Section B, you will be allowed to work with the apparatus for a maximum of 1 hour.
You are expected to record all your observations as soon as these observations are made.
An account of the method of carrying out the experiments is **not** required.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

This document consists of **8** printed pages and **4** blank pages.

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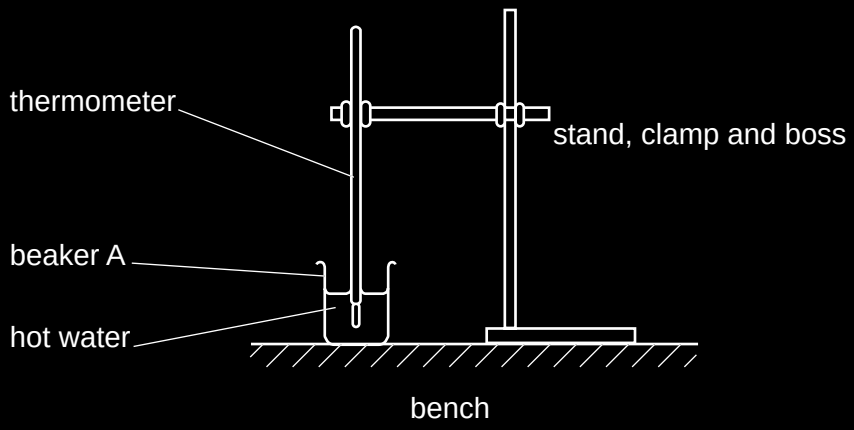


Fig. 2.1

Stir the water. Record the maximum temperature θ_2 shown on the thermometer before the temperature starts to fall.

θ_2 = ...

(v) Immediately pour the 50 cm^3 of water from the measuring cylinder into beaker A. Stir the mixture. The reading on the thermometer will fall quickly at first and then at a slower rate. As soon as it starts to fall at the slower rate, record the temperature θ_3 of the mixture.

θ_3 = ...

[3]

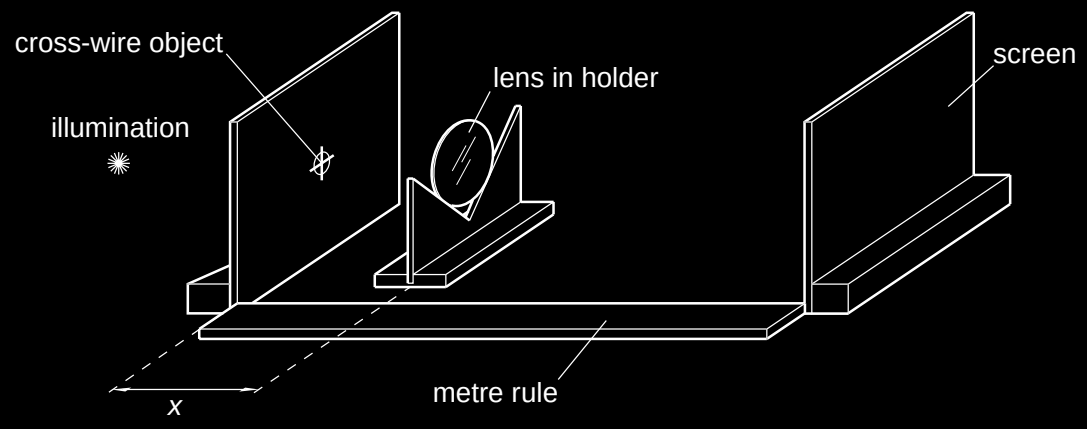


Fig. 3.1

Adjust the position of the lens until a magnified, focussed image of the object is formed on the screen. Measure the distance *x* between the object and the lens.

x = ... [1]

(b) **(i)** State, using your observations, whether the image is upright or inverted.

.....

(ii) Describe the observations that you made.

.....

.....

[1]

(c) Keeping the object and the screen in the same position, move the lens until a diminished, focussed image of the object is formed on the screen. Measure the distance *y* between the object and the lens.

y = ... [1]

(d) Calculate the focal length *f* of the lens using;

$$f = \frac{D^2 - d^2}{4D}$$

where *D* is the distance between the object and the screen
and $d = y - x$.

f = ... [2]

[Total: 5]

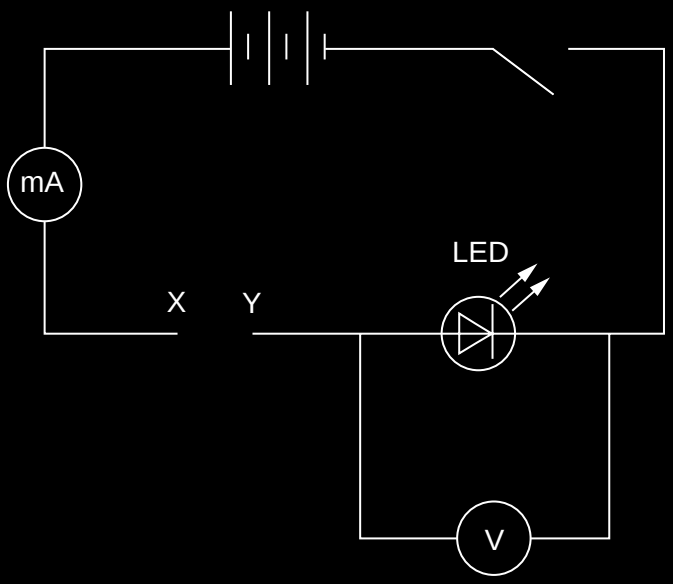


Fig. 4.1

(a) **(i)** Connect a resistor of resistance R = 270 Ω in the gap between points X and Y.

(ii) Complete the circuit by closing the switch. Measure the current I in the circuit and the potential difference V across the light emitting diode (LED). Open the switch.

I = ...

V = ...
[2]

(b) Calculate the resistance R_{LED} of the LED by using

$$R_{LED} = \frac{V}{I}$$

R_{LED} = ... [2]

